Available online at www.sciencedirect.com
SCIENCE DIRECT®
European Journal of Operational Research 173 (2006) 488–507
EUROPEAN JOURNAL OF OPERATIONAL RESEARCH
www.elsevier.com/locate/ejor

Discrete Optimization

Mathematical programming models and algorithms for a class–faculty assignment problem

Salem M. Al-Yakoob [a,*], Hanif D. Sherali [b,1]

[a] Department of Mathematics and Computer Science, College of Science, Kuwait University, P.O. Box 5969, Safat 13060, Kuwait
[b] Grado Department of Industrial and Systems Engineering (0118), Virginia Polytechnic Institute and State University, Blacksburg, VA 24061, United States

Received 11 May 2004; accepted 28 January 2005
Available online 30 March 2005

Abstract

This paper presents mathematical programming models for assigning faculty members to classes including, among typical academic class scheduling issues, certain specialized central policies at Kuwait University. The time-slots for classes are initially assumed to be given and an integer programming model (CFAM) is constructed to solve the resulting problem, which aims to minimize the individual and collective dissatisfaction of faculty members in a fair fashion, where dissatisfaction is measured by a function of the assignment of faculty members to time-slots and specific classes. In order to enhance the quality of results obtained in practice, the model is modified (ECFAM) so that the time-slots for the classes can be changed, however, with restrictions related to efficient facility utilization and permitting an administratively regulated maximum number of changes. Gender-based modeling considerations are also introduced in order to maintain desirable class offering patterns. Computational results are presented based on solving the models directly by the CPLEX-MIP (version 7.5) package and also using a specialized LP-based heuristic. The faculty schedules generated via the proposed approach based on a number of case studies related to the Department of Mathematics and Computer Science at Kuwait University reveal that this approach yields improved schedules in terms of fairness and enhanced satisfaction levels among faculty members.
© 2005 Elsevier B.V. All rights reserved.

Keywords: Timetabling; Class scheduling; Mixed-integer programming; Employee scheduling; Gender policies

* Corresponding author. Tel.: +965 4811188 5371; fax: +965 4817201.
E-mail addresses: salem@al-yakoob.com (S.M. Al-Yakoob), hanifs@vt.edu (H.D. Sherali).
[1] Tel.: +1 540 231 5474; fax: +1 540 231 3322.

0377-2217/$ - see front matter © 2005 Elsevier B.V. All rights reserved.
doi:10.1016/j.ejor.2005.01.052

1. Introduction

1.1. Motivation and problem specifications

A typical class-scheduling problem is concerned with the assignment of classes to appropriate faculty members, suitable classrooms, and available time-slots. Many large universities face this formidable multidimensional assignment task. The focus of the present research effort is directed to the class-faculty assignment problem at Kuwait University (KU), which to a large extent, manually handles this multidimensional task every semester. The generated class schedules are often accompanied by numerous problems pertaining to an inefficient utilization of facility resources, and biased faculty-class assignments. Class conflicts, dissatisfaction among faculty members, and parking and traffic congestions are common problems encountered each semester. Recently, this task has been further complicated by (1) a new gender-based policy imposed by the national assembly (Law 24 of year 1999), and (2) the Office of Registration's new policy related to fixing the timetable except for a maximum of 15% permitted changes. A full implementation of the gender segregation policy is likely to require more facility and human resources because, for example, certain single section classes must be partitioned into two different class sections, each of which requires its own classroom and faculty member. The 15% change policy imposed by the Office of Registration cannot be effectively implemented based on the current inefficient scheduling approaches; rather, it must be based on scheduling procedures that provide for flexible class and time offering patterns, and that ensure an efficient utilization of facility and human resources.

One of the authors has served in the class scheduling committee in the Mathematics and Computer Science Department during the period 1997–2002. He realized that many of the shortcomings and inefficiencies of the current scheduling approaches are attributed to (1) a lack of coordination between departments that share common facility resources, thereby leading to an inefficient utilization of facility resources and producing many problems related to class conflicts, unavailability of proper classrooms, and traffic and parking congestions, and (2) preferences of faculty members for time-slots that tend to cluster classes during certain periods of the day. Accordingly, we have proposed a two-stage approach for the academic class scheduling and timetabling problem at Kuwait University (KU). Stage I is concerned with designing an efficient class timetable that provides flexible class and time offering patterns, while considering the available facility resources such as classrooms, laboratories, and parking and traffic facilities. A companion paper [2] addresses this Stage I problem. The present Stage II paper is concerned with assigning faculty members to classes for individual departments (the time-slots for classes being now fixed as determined in Stage I, except that a maximum of 15% changes are permitted in rescheduling classes, as newly adopted by the Office of Registration). We have adopted such a two-stage approach, as opposed to using a single composite model that incorporates both stages I and II for the following reasons. First, a model that combines both stages is likely to be far too complex to solve effectively. Second, faculty members' preferences for specific time-slots and classes is a secondary issue when there is a scarcity of resources, and hence, an efficient utilization of such resources is of paramount importance, as in the case of KU. Third, a timetable that is produced by an aggregate model might not be satisfactory to faculty members in the following year when the mix of faculty and their preferences change. Moreover, the imposed Office of Registration's limit on the number of permitted changes in the timetable is not so conveniently applied to such a complex composite model. Furthermore, several objective criteria are of importance when designing class timetabling systems, including: (a) maximizing the utilization of facility and human resources, (b) ensuring a balance of class and time offering patterns, and (c) minimizing individual and collective dissatisfaction levels of faculty members, being measured in terms of desired classes and time-slots. The priorities of such objectives to be achieved by any scheduling system are usually determined by decision-makers. Some universities need to focus on the scarcity of resources and on the increasing number of students, while others are more concerned with the class offering patterns. Based on these

considerations and our discussions with administrators, the two-stage approach was deemed to be most appropriate for KU.

A typical scheduling committee manually generates the faculty-class assignment each semester. Although such scheduling committees spend an inordinate effort deciding upon who teaches what and when, the generated faculty schedules and the job satisfaction levels they provide are far from being desirable. It is therefore imperative to utilize more efficient approaches to handle the highly combinatorial nature of the problem.

1.2. Related research

The faculty-class assignment problem presented in this paper can be viewed as an employee-scheduling problem. In general, employee scheduling problems are concerned with the assignment of employees to specific shifts in order to satisfy certain types of work requirements based on employees' qualifications [3]. In the specific problem that is the focus of this paper, employees are faculty members, shifts are the specified time-slots, work requirements are the classes to be offered, and employees' qualifications correspond to the ability of each faculty member to teach different classes. The satisfaction of faculty members' preferences is an essential objective that our proposed model aims to achieve to the extent possible. Many models related to employee preferences have been addressed in the literature. Nanda and Browne [17] present detailed discussions and simple mathematical programming models related to employees' preferences for a number of employee scheduling problems. Yura [21] formulates an employee scheduling problem as a linear goal program that incorporates employees' preferences by considering different scenarios of satisfying such preferences with respect to off-days and overtime assignments. Tackling employees' preferences within a goal programming context has also been considered by Ozkarahan and Bailey [18], who designed a decision support system for a nurse scheduling problem. The objective function of this goal program is composed of nurses' preferences, labor costs, and penalties for deviating from certain staffing goals. Another decision support system based on similar goal programming concepts has been developed by Ozkarahan [19].

Academic class scheduling problems often admit multidimensional assignment formulations, whose solutions depend to a large extent upon intricacies of the scheduling environment and the size of the problem in terms of the number of variables and constraints. A wide spectrum of related multidimensional optimization problems have been investigated. Two-dimensional assignment problems, in particular, have been extensively studied in the literature, and fast and efficient solution algorithms have been developed for certain classes of these problems. Three-dimensional assignment problems, although notorious for their complexity, have also drawn considerable attention due to the fact that many real-world problems admit such formulations (see, e.g., Gilbert and Hofstra [14]). Four- and higher-dimensional assignment problems normally require heuristic procedures and massive computing power in order to derive good quality feasible solutions.

Class scheduling and timetabling problems have been widely addressed in the literature, and many mathematical models have been developed to tackle such problems. However, due to the specific nature of each individual problem, there is no universal model that can be utilized to solve all such problems. Among many such specialized problem treatments that are relevant to this paper, we refer the reader to Astratian and de Werra [4], Astratian and Mirumian [5], Aubin and Ferland [6], Costa [7], de Werra [8,9], Dowsland [10], Fahrion and Dollansky [11], Ferland and Fleurent [12], Giannikos et al. [13], Gotlieb [15], Lawrie [16], and Tripathy [20]. A more detailed literature review on class scheduling problems is presented in the companion paper [2].

Although assignment and academic scheduling problems are somewhat similar to the problem addressed in this paper, none of these fully considers all the aspects that are pertinent to our problem. In particular, gender related policies and various restrictions pertaining to teaching different classes and teaching classes consecutively are addressed and incorporated into the proposed model. Furthermore, faculty members'

preferences in terms of specific classes and time-slots are considered in order to minimize the overall dissatisfaction level of employees. For the sake of fairness, certain equity measures among faculty teaching similar loads are also incorporated within the model. The proposed model additionally generates class schedules that conform with the new policy adopted by the Office of Registration, which permits at most 15% changes based on the class schedule of the previous year. Note that many of these novel modeling considerations and equity concepts are transportable to related problems faced at other universities.

The remainder of this paper is organized as follows. The next section addresses basic guidelines and considerations for faculty members' preferences with respect to classes and time-slots, and Section 3 provides a description of the overall problem. A mixed-integer programming model for the problem is formulated in Section 4, and a modified version of this model that allows limited time changes for classes while maintaining certain essential features of the prescribed problem is described in Section 5. Section 6 develops modeling strategies related to gender policy issues. Solution algorithms and computational results are presented in Section 7. Test instances based on real class and faculty data for three academic years are solved directly by the CPLEX-MIP Package and also via the specialized developed algorithms. Finally, Section 8 concludes the paper with a summary and some implementation recommendations.

2. Preferences of faculty members

Faculty members' preferences for teaching selected classes during specific time-slots are an integral component of any efficient class scheduling and timetabling system. It is a rather complex task, and sometimes impossible, to assign classes to faculty members in a way that perfectly meets every individual faculty members' preferences for specific classes and time-slots. Moreover, a manual assignment of classes to faculty members is often biased and leads to discrepancies in satisfaction levels among faculty members. The need for developing detailed guidelines that govern faculty preferences, and the importance of developing efficient approaches that equitably assign classes to faculty members, intensifies in relatively large departments where faculty members compete for certain classes and time-slots. For example, the Mathematics and Computer Science Department at Kuwait University assigns around 120 classes to about 50 faculty members, and the foregoing concerns are very evident in this department. The remainder of this section is concerned with guidelines for faculty members' preferences for specific classes and time-slots. Note that most classes are for three credits and that each class is offered either on Saturday, Monday, and Wednesday or on Sunday and Tuesday. Let SMW = {Saturday, Monday, Wednesday} and let ST = {Sunday, Tuesday}. The time-slots for the SMW and ST classes are, respectively, of 50 and 75 minutes duration.

2.1. Restrictions on the consideration of faculty members' preferences

The consideration of faculty members' preferences are often governed by predetermined guidelines that are developed over a period of time to ensure fairness and appropriateness in the process of assigning classes to faculty members. These guidelines are dynamic in nature and vary from one university to another, and sometimes, even vary within departments of one university due to the peculiarities of individual departments. The following related restrictions are imposed in our class-faculty assignment models.

(A) If the teaching load of a faculty member is two classes, then his/her classes must be different, i.e., in this case, a faculty member cannot teach two sections of the same class.
(B) If the teaching load of a faculty member is three classes, then the faculty member has the choice of teaching (at most) two sections of one class and one section of another, or of teaching three different classes.

(C) A faculty member may teach at most two classes consecutively, if so desired.
(D) A faculty member may not teach a single section class for three consecutive semesters. This condition is needed to avoid the monopolization of certain classes by some faculty members, which is sometimes also undesirable from the students' viewpoint. However, this condition is relaxed in some special cases, as for example, if no other faculty member is willing or qualified to teach a class that has been taught consecutively by the same faculty member.
(F) Certain higher-level classes must be taught only by faculty members who are specialized in the related field.

The above restrictions may not be suitable for small sized departments. However, these restrictions are essential for departments that offer many multiple sections of lower-level classes, such as for example, the Department of Mathematics and Computer Science at KU that typically offers 16, 14, and 9 sections of Calculus I, II, and III, respectively.

2.2. Compiling faculty members' preferences

Keeping the foregoing restrictions in mind, our approach requires every faculty member to fill out a questionnaire pertaining to his/her preferences for specific classes and time-slots, based on the offered classes and their fixed time-slots. The questionnaire is basically composed of three parts as follows.

2.3. General information

This part records preferences of faculty members regarding teaching three different classes and teaching classes consecutively.

2.4. Time-slot preferences

Assume that there are $A + B$ time-slots during which classes are held, where A denotes the number of available time-slots on SMW and B denotes the number of available time-slots on ST. Even though in the current scheduling practice $A + B = 15$ with $A = 9$, $B = 6$ based on 8 a.m. to 5 p.m. classes, we allow $A + B$ to take higher values (with classes extending up to 8:00 p.m.) in order to examine the impact of having different values of A and B on the class timetable. Each faculty member is instructed to select 10 time-slots and to assign a number from the set $\{1, 2, \ldots, 10\}$ to each time-slot. No two time-slots can be assigned the same number, and the lowest assigned number indicates the highest preference. Hence, the time-slot associated with number "1" represents the first time-slot preference, and that associated with the number "10" represents the last time-slot preference. (It is worth mentioning that faculty members involved in administrative college duties, e.g., serving as a Dean of a College or a Chairman of a Department, often have tight schedules, and therefore, such faculty members should be allowed to provide a smaller list of time-slot preferences than other faculty members.)

2.5. List of classes

Each faculty member is required to list seven different classes, the first of which represents the highest preference and the last of which represents the lowest preference. We associate the number "1" with the first choice, "2" with the second choice, and so on. At least three classes must come from freshman and sophomore levels, with at least one class from the freshman level and at least one class from the sophomore level.

3. Problem description

In this section, the notation, and a faculty dissatisfaction cost function are defined in order to formulate the desired integer programming models in Sections 4 and 5. For convenience in reading, a glossary of notation is provided in Appendix A.

3.1. Problem notation

Let $t = 1, \ldots, A$ index the SMW time-slots, and let $t = A + 1, \ldots, A + B$ index the ST time-slots. Let $T_{\mathrm{SMW}} = \{1, \ldots, A\}$, $T_{\mathrm{ST}} = \{A + 1, \ldots, A + B\}$, and $T = \{1, \ldots, A + B\}$.

Let $c = 1, \ldots, C$ denote the different classes that are offered during the specified term. The number of sections of each class and the time-slots associated with each class are known a priori. Let $F_{c,t}$, $M_{c,t}$, and $J_{c,t}$ respectively denote the number of female, male, and joint sections of class c offered during time-slot t. The total numbers of female, male, and joint sections of class c are respectively denoted by F_c, M_c, and J_c. Hence, $F_c = \sum_t F_{c,t}, M_c = \sum_t M_{c,t}$, and $J_c = \sum_t J_{c,t}$.

Let $E_c = (F_c, M_c, J_c)$. Let $O_{c,t}$ denote the total number of sections of class c offered during time-slot t, regardless of gender, being equal to $F_{c,t} + M_{c,t} + J_{c,t}$. Note that $O_{c,t} = 0$ whenever no section of class c is offered during time-slot t. The total number of sections of class c offered during the different time-slots is denoted by O_c, being equal to $\sum_t O_{c,t}$. The total number of sections of classes offered during time-slot t is denoted by O_t, being equal to $\sum_c O_{c,t}$. Note that if no gender considerations are treated in the problem, then we may only deal with $O_{c,t}$ and disregard $F_{c,t}$, $M_{c,t}$, and $J_{c,t}$. The average number of students that are expected to register for a class c (i.e., the class size) is known beforehand based on information related to historical statistics and students' major sheets. Let $s_1, s_2, \ldots, s_m$ denote all possible designated class sizes, and accordingly, partition $\{1, \ldots, C\}$ into mutually exclusive sets $S_1, S_2, \ldots, S_m$, where for $j = 1, \ldots, m$, S_j is composed of all classes having class size s_j. Let R_{t,s_j} denote the number of classes, each of size s_j, that are held during time-slot t.

Let $f = 1, \ldots, F$ denote the faculty members that are available for the term, and for each f, let L^f denote the class-load for faculty member f. This class-load represents the maximum number of classes that can be possibly assigned to a given faculty member, and it is one, two, or three. The precise value of L^f is determined based on well-defined university guidelines that consider the rank of the faculty member, the administrative duties assigned to the faculty member, and research commitments.

Let L_1, L_2, and L_3 denote the sets of faculty members teaching one, two, and three classes, respectively. Partition L_2 into two subsets, L_{21} and L_{22}, the former being the set of faculty members who are willing to teach two classes consecutively, and the latter being the complement of L_{21} in L_2. Likewise, L_3 is partitioned in a similar fashion into two subsets, L_{31} and L_{32}. Also, partition L_3 into two other subsets, L_{33} and L_{34}, the former being composed of faculty members who are willing to teach three different classes, and the latter being the complement of L_{33} in L_3. Note that $L_3 = L_{31} \cup L_{32} = L_{33} \cup L_{34}$. Also, for $i \in \{1, 2, 3\}$, define $Q_i = \{(f_1, f_2)$: where both f_1 and f_2 belong to L_i with $f_1 < f_2\}$. Let $L = L_1 \cup L_2 \cup L_3$ and $Q = Q_1 \cup Q_2 \cup Q_3$. Let $MJ \subseteq L$ be the set of faculty members who are willing to teach male and joint sections of offered classes.

Let T_c be the set of time-slots during which sections of class c are offered, and let A_c be the set of faculty members that can teach class c. Let C^f be the set of classes that can be taught by faculty member f, some of which are selected by this faculty member, while others are included by the administration, but with a suitable high objective penalty function to indicate any undesirability on the part of the faculty member to teach this class. Let C_t be the set of classes that are offered during time-slot t.

3.2. Faculty dissatisfaction cost function

Based on the responses obtained from the questionnaire discussed in Section 2.2, a cost function representing the dissatisfaction level of each faculty member is defined in this section.

Let $I = \{(c,f,t): c \in \{1,\ldots,C\}, f \in A_c, \text{ and } t \in T_c\}$. Hence, the set I is restricted to incorporate only valid combinations of (c,f,t). Two instances, for example, of invalid combinations of (c,f,t) are if c is a class that cannot be taught by faculty member f, or if c is not offered during time-slot t. Note that the set I may also exclude some other specific valid combinations of (c,f,t), if so desired by the scheduler.

Let N^+ denote the set of positive integers. Define a *dissatisfaction function* $DC: I \to N^+$ by $\mathrm{DC}(c,f,t) = \kappa(c,f) + \lambda(f,t)$, where $\kappa(c,f)$ and $\lambda(f,t)$ are positive integer numbers representing the respective priorities associated with class c and time-slot t as indicated in the questionnaire filled out by faculty member f. (Note that relative high penalties are ascribed to unlisted values of c and t.)

4. Model formulation

4.1. An integer programming model

Consider the following binary decision variables.

Let

$$X_{c,f,t} = \begin{cases} 1 & \text{if a section of class } c \text{ is taught by faculty member } f \text{ during time-slot } t, \\ 0 & \text{otherwise,} \end{cases}$$

where the indices $(c,f,t) \in I$ are defined in Section 3.2.

For a faculty member $f \in L_{34}$, we define the following binary decision variables.

$$H_{c,f} = \begin{cases} 1 & \text{if faculty member } f \in A_c \text{ teaches at least one section of class } c, \\ 0 & \text{otherwise.} \end{cases}$$

Given two faculty members f_1 and f_2 such that $(f_1,f_2) \in Q$, the integer variable Y_{f_1,f_2} represents the absolute difference between the dissatisfaction levels of faculty members f_1 and f_2. (This variable is used to model equity considerations between faculty members having the same teaching load.)

The prescribed Course-Faculty Assignment Model (CFAM) can then be stated as follows. (All indices are assumed to take on only their respective relevant values.)

CFAM

$$\text{Minimize} \quad \sum_{(c,f,t)\in I} DC(c,f,t)X_{c,f,t} + \mu \sum_{(f_1,f_2)\in Q} Y_{f_1,f_2}$$

subject to

$$(\mathrm{C}_1) \quad \sum_{f\in A_c} X_{c,f,t} = O_{c,t} \quad \forall c, t \in T_c,$$

$$(\mathrm{C}_2) \quad \sum_{c\in C^f} \sum_{t\in T_c} X_{c,f,t} \leqslant L^f \quad \forall f,$$

$$(\mathrm{C}_3) \quad \sum_{c\in C^f\cap C_t} X_{c,f,t} \leqslant 1 \quad \forall t \quad \text{and} \quad f \in L_2 \cup L_3,$$

$$(\mathrm{C}_4) \quad \sum_{c\in C^f\cap C_t} X_{c,f,t} + \sum_{c\in C^f\cap C_{t+1}} X_{c,f,t+1} \leqslant 1 \quad \forall t = 1,\ldots,A-1,A+1,\ldots,A+B-1 \quad \text{and}$$
$$f \in L_{22} \cup L_{32},$$

$$(\mathrm{C}_5)\quad \sum_{c\in C^f\cap C_t} X_{c,f,t} + \sum_{c\in C^f\cap C_{t+1}} X_{c,f,t+1} + \sum_{c\in C^f\cap C_{t+2}} X_{c,f,t+2} \leqslant 2$$
$$\forall t = 1,\ldots,A-2, A+1,\ldots,A+B-2 \quad \text{and} \quad f\in L_{31},$$

$$(\mathrm{C}_6)\quad \sum_{t\in T_c} X_{c,f,t} \leqslant 1 \quad \forall c, f\in L_2\cap A_c,$$

$$(\mathrm{C}_7)\quad \sum_{t\in T_c} X_{c,f,t} \leqslant 2 \quad \forall c, f\in L_3\cap A_c,$$

$$(\mathrm{C}_{8.1})\quad H_{c,f} \geqslant X_{c,f,t} \quad \forall c, f\in L_{34}\cap A_c, t\in T_c,$$

$$(\mathrm{C}_{8.2})\quad H_{c_1,f} + H_{c_2,f} + H_{c_3,f} \leqslant 2 \quad \forall c_1 < c_2 < c_3 \text{ in } C^f \text{ with } f\in L_{34},$$

$$(\mathrm{C}_{9.1})\quad Y_{f_1,f_2} \geqslant \sum_{c\in C^{f_1}}\sum_{t\in T_c} DC(c,f_1,t)X_{c,f_1,t} - \sum_{c\in C^{f_2}}\sum_{t\in T_c} DC(c,f_2,t)X_{c,f_2,t,} \quad \forall (f_1,f_2)\in Q,$$

$$(\mathrm{C}_{9.2})\quad Y_{f_1,f_2} \geqslant \sum_{c\in C^{f_2}}\sum_{t\in T_c} DC(c,f_2,t)X_{c,f_2,t,} - \sum_{c\in C^{f_1}}\sum_{t\in T_c} DC(c,f_1,t)X_{c,f_1,t} \quad \forall (f_1,f_2)\in Q,$$

$$X_{c,f,t}\in\{0,1\} \quad \forall (c,f,t)\in I \text{ and equal to zero otherwise,}$$

$$Y_{f_1,f_2} \geqslant 0 \quad \forall (f_1,f_2)\in Q \text{ and equal to zero otherwise,}$$

$$H_{c,f}\in[0,1] \quad \forall c \quad \text{and} \quad f\in L_{34}\cap A_c.$$

4.1.1. Objective function

The objective function of the problem seeks to minimize the total dissatisfaction and inequity costs associated with assigning faculty members to classes. The total dissatisfaction cost is given by the first part of the objective function, and the sum of differences in dissatisfaction levels between faculty members having identical teaching loads is given by the second part of the objective function. A weight factor μ is associated with the second objective function term to reflect its relative priority with respect to the first term.

4.1.2. Constraints

Constraint (C_1) guarantees that all offered sections of each class are assigned to proper faculty members. Constraint (C_2) asserts that the total number of classes assigned to each faculty member does not exceed his/her known class-load, while constraint (C_3), along with (C_2) for the case $f\in L_1$, asserts that at most one class may be assigned to a given faculty member during any specified time-slot. Restrictions regarding teaching classes consecutively are enforced by constraints (C_4) and (C_5). Constraint (C_4) prohibits assigning two classes consecutively to a faculty member if he/she does not desire this, and Constraint (C_5) assures that no faculty member teaches three classes consecutively. Recall that a faculty member having $L^f = 2$ must be assigned two different classes and a faculty member having $L^f = 3$ may teach at most two sections of the same class. These restrictions are enforced by constraints (C_6) and (C_7), respectively. Furthermore, for a class c and faculty member $f\in L_{34}\cap A_c$, the variable $H_{c,f}$ in Constraint ($\mathrm{C}_{8.1}$) receives a value of 1 if faculty member f teaches at least one section of class c and Constraint ($\mathrm{C}_{8.2}$) fulfills the request of this faculty member regarding not teaching three different classes. Note that given any binary (feasible) values of the X-variables, since the satisfaction of ($\mathrm{C}_{8.2}$) prefers lower values for H-variables subject to ($\mathrm{C}_{8.1}$), the integrality of the H-variables in an alternative optimal solution is automatically ensured even if these variables are

defined in the continuous interval $[0,1]$. Constraints ($C_{9.1}$) and ($C_{9.2}$) give the absolute difference between dissatisfaction levels for pairs of faculty members having the same class-load. Note that the integrality of the variable Y_{f_1,f_2} is automatically assured by constraints ($C_{9.1}$) and ($C_{9.2}$) and the objective function.

Remark 1. Suppose that $\kappa = \lambda = 1$. If the class-load for faculty member f is $L^f = 2$, then the ideal assignment for faculty member f occurs, if his/her class timetable is composed of the first two class preferences being held during the first two time-slot preferences. In this case, $\sum_{c\in C^f}\sum_{t\in T_c} DC(c,f,t)X_{c,f,t} = 6$. On the other hand, if the class-load for faculty member f is $L^f = 3$, then the ideal assignment for faculty member f yields $\sum_{c\in C^f}\sum_{t\in T_c} DC(c,f,t)X_{c,f,t} = 10, 11$, or 12, depending on whether two sections of the first or second preference class are offered, or if all three classes taught are distinct. For $L^f = 1$, the ideal assignment for faculty member f yields $\sum_{c\in C^f}\sum_{t\in T_c} DC(c,f,t)X_{c,f,t} = 2$. Accordingly, we can assess the actual dissatisfaction level of any faculty member f for a given class assignment based on the incremental value of $\sum_{c\in C^f}\sum_{t\in T_c} DC(c,f,t)X_{c,f,t}$ beyond the foregoing associated minimal or ideal levels.

Remark 2. The second term in the objective function and constraints ($C_{9.1}$) and ($C_{9.2}$) are designed to minimize differences in dissatisfaction levels between faculty members having the same teaching load. The positive weight given by μ indicates the relative effect of enforcing fairness among faculty members versus minimizing the total system dissatisfaction. The fairness issue can be further emphasized, if so desired, by introducing a hard constraint that ensures a specified maximum level of dissatisfaction difference between faculty members having the same teaching load. This may be accomplished by imposing upper bounds $Y_{f_1,f_2} \leqslant UD_l \ \forall (f_1,f_2) \in Q_l$, where $l = 1,2,3$.

5. Enhancing satisfaction of faculty members

In this section, we present another model, denoted by ECFAM, for the class-faculty assignment problem described in Section 1. This model is basically derived from Model CFAM and it aims to further enhance the satisfaction of faculty members' preferences by permitting classes to be offered during time-slots other than those already specified. Accordingly, Constraint (C_1) must be dropped to allow changes in offering patterns that can be accomplished by allowing certain X-variables for which $(c,f,t) \notin I$ to take a value of 1. However, the total number of sections of class c that need to be offered remains O_c, which can be enforced by the following constraint.

$$(\mathrm{EC}_{10}) \quad \sum_t \sum_{f\in A_c} X_{c,f,t} = O_c \quad \forall c.$$

In the remainder of this section, we present additional constraints that will be used along with modified versions of constraints (C_2), ..., ($C_{9.2}$), which are respectively denoted by (EC_2), ..., ($EC_{9.2}$), to formulate Model ECFAM. Note that changes in time-slots of sections of offered classes should retain certain essential features of the class timetable obtained in Stage I, as discussed in Section 1.1. These features attempt to efficiently utilize facility resources and minimize class conflicts. Accordingly, while maintaining such features, Model CFAM could be modified to reflect allowing restricted changes in time-slots. These modifications are discussed next.

5.1. Utilization of facility resources and offering pattern considerations

Certain constraints could be added to the set of constraints of Model CFAM in order to maintain an efficient utilization of facility resources and to reduce class conflicts. These constraints are formulated and explained below.

5.2. Retaining the same number of classes having the same class size during any given time-slot

For any time-slot t, the total number of classes having a specified class size s_j as given by R_{t,s_j}, say, should not change in order to ensure the availability of proper classrooms. This is enforced by the following constraint:

$$(\mathrm{EC}_{11}) \quad \sum_{c \in S_j} \sum_{f \in A_c} X_{c,f,t} \leqslant R_{t,s_j} \quad \forall t, j = 1, \ldots, m.$$

5.3. Multiple-section classes

Multiple-section classes should be spread over the available time-slots to provide students with more alternatives. Hence, let c be a multiple section class and let D_c be a targeted maximum number of sections that may be offered during any time-slot t. Moreover, let $Z_{t,c}$ be a nonnegative integer variable that represent the number of offered sections that exceed the targeted level D_c during time-slot t, being defined by

$$(\mathrm{EC}_{12}) \; Z_{c,t} \geqslant \left(\sum_{f \in A_c} X_{c,f,t} - D_c \right) \quad \forall c, t.$$

If γ_c is some appropriate penalty ascribed per section offered beyond the level D_c during any time-slot, then the corresponding objective function penalty term is given by $\sum_c \sum_t \gamma_c Z_{c,t}$. Note that since $Z_{c,t}$ is nonnegative for any class c and time-slot t, the penalty term $\gamma_c \, Z_{c,t}$ vanishes as long as the number of offered sections does not exceed D_c.

5.4. Minimizing class conflicts

Certain single-section classes should not be offered during the same time-slot so that a student could simultaneously take any two of such classes. For this reason, special sets of classes are constructed and a penalty is associated with each set to reflect the importance of not offering classes from this set during the same time-slot. A higher penalty value indicates a greater undesirability for offering any two classes of the set during the same time-slot. Let $\Lambda_1, \ldots, \Lambda_n$ denote such special sets of classes, and let λ_j be the penalty associated with set Λ_j. Let $P_{t,j}$ be a nonnegative integer variable representing the number of class conflicts during time-slot t for classes that belong to the set Λ_j. The value of this variable is determined by the following constraint.

$$(\mathrm{EC}_{13}) \; P_{t,j} \geqslant \left(\sum_{c \in \Lambda_j} \sum_{f \in A_c} X_{c,f,t} - 1 \right), \quad \forall t, j = 1, \ldots, n.$$

The corresponding objective function penalty term is given by $\sum_t \sum_j \lambda_j P_{t,j}$.

5.5. Enforcing a maximum number of changes

The maximum number of allowable time-slot changes for sections of offered classes is enforced by the following three constraints, where $W_{c,t}$ is a nonnegative integer variable that records changes in the total number of sections of class c that are offered during time-slot t, and TAC is the total number of allowable changes.

$$(\mathrm{EC}_{14.1}) \quad W_{c,t} \geqslant \left(\sum_{f \in A_c} X_{c,f,t} - O_{c,t}\right) \quad \forall c, t,$$

$$(\mathrm{EC}_{14.2}) \quad W_{c,t} \geqslant \left(O_{c,t} - \sum_{f \in A_c} X_{c,f,t}\right) \quad \forall c, t,$$

$$(\mathrm{EC}_{14.3}) \quad \sum_c \sum_t W_{c,t} \leqslant \mathrm{TAC}.$$

The number of sections for a given class per time-slot is now allowed to change in order to enhance individual and overall satisfaction of faculty members. This requires the modification of the set I as defined in Section 3.2 to incorporate more combinations of (c,f,t). Accordingly, let $I^E = \{(c,f,t) : \forall c \in \{1,\ldots,C\}, f \in A_c, \text{ and } t \in T\}$ replace I, where I^E is composed of all possible combinations (c,f,t) except those for which a faculty member is assigned to a class that he/she cannot teach. The modified version of Model CFAM can now be stated as follows.

ECFAM

$$\text{Minimize} \quad \sum_{(c,f,t) \in I^E} DC(c,f,t) X_{c,f,t} + \mu_1 \sum_{(f_1,f_2) \in Q} Y_{f_1,f_2} + \mu_2 \sum_c \sum_t \gamma_c Z_{c,t} + \mu_3 \sum_t \sum_j \lambda_j P_{t,j}$$

subject to

$$(\mathrm{EC}_2) \quad \sum_{c \in C^f} \sum_t X_{c,f,t} \leqslant L^f \quad \forall f,$$

$$(\mathrm{EC}_3) \quad \sum_{c \in C^f} X_{c,f,t} \leqslant 1 \quad \forall t \quad \text{and} \quad f \in L_2 \cup L_3,$$

$$(\mathrm{EC}_4) \quad \sum_{c \in C^f} X_{c,f,t} + \sum_{c \in C^f} X_{c,f,t+1} \leqslant 1, \quad \forall t = 1,\ldots,A-1, A+1,\ldots,A+B-1 \quad \text{and} \quad f \in L_{22} \cup L_{32},$$

$$(\mathrm{EC}_5) \quad \sum_{c \in C^f} X_{c,f,t} + \sum_{c \in C^f} X_{c,f,t+1} + \sum_{c \in C^f} X_{c,f,t+2} \leqslant 2,$$
$$\forall t = 1,\ldots,A-2, A+1,\ldots,A+B-2 \quad \text{and} \quad f \in L_{31},$$

$$(\mathrm{EC}_6) \quad \sum_t X_{c,f,t} \leqslant 1 \quad \forall c, f \in L_2 \cap A_c,$$

$$(\mathrm{EC}_7) \quad \sum_t X_{c,f,t} \leqslant 2 \quad \forall c, f \in L_3 \cap A_c,$$

$$(\mathrm{EC}_{8.1}) \quad H_{c,f} \geqslant X_{c,f,t} \quad \forall c, f \in L_{34} \cap A_c, t,$$

$$(\mathrm{EC}_{8.2}) \quad H_{c_1,f} + H_{c_2,f} + H_{c_3,f} \leqslant 2 \ \forall c_1 < c_2 < c_3 \quad \text{in } C^f \quad \text{with } f \in L_{34},$$

$$(\mathrm{EC}_{9.1}) \quad Y_{f_1,f_2} \geqslant \sum_{c \in C^{f_1}} \sum_t DC(c,f_1,t) X_{c,f_1,t} - \sum_{c \in C^{f_2}} \sum_t DC(c,f_2,t) X_{c,f_2,t} \quad \forall (f_1,f_2) \in Q,$$

$$(\mathrm{EC}_{9.2}) \quad Y_{f_1,f_2} \geqslant \sum_{c \in C^{f_2}} \sum_t DC(c,f_2,t) X_{c,f_2,t} - \sum_{c \in C^{f_1}} \sum_t DC(c,f_1,t) X_{c,f_1,t} \quad \forall (f_1,f_2) \in Q,$$

$$(\mathrm{EC}_{10}) \quad \sum_t \sum_{f \in A_c} X_{c,f,t} = O_c \quad \forall c,$$

$$(\mathrm{EC}_{11}) \quad \sum_{c \in S_j} \sum_{f \in A_c} X_{c,f,t} \leqslant R_{t,s_j} \quad \forall t, j = 1, \ldots, m,$$

$$(\mathrm{EC}_{12}) \quad Z_{c,t} \geqslant \left(\sum_{f \in A_c} X_{c,f,t} - D_c \right) \quad \forall c, t,$$

$$(\mathrm{EC}_{13}) \quad P_{t,j} \geqslant \left(\sum_{c \in A_j} \sum_{f \in A_c} X_{c,f,t} - 1 \right) \quad \forall t, j = 1, \ldots, n,$$

$$(\mathrm{EC}_{14.1}) \quad W_{c,t} \geqslant \left(\sum_{f \in A_c} X_{c,f,t} - O_{c,t} \right) \quad \forall c, t,$$

$$(\mathrm{EC}_{14.2}) \quad W_{c,t} \geqslant \left(O_{c,t} - \sum_{f \in A_c} X_{c,f,t} \right) \quad \forall c, t,$$

$$(\mathrm{EC}_{14.3}) \quad \sum_c \sum_t W_{c,t} \leqslant \mathrm{TAC},$$

$X_{c,f,t} \in \{0,1\}$ $\forall (c,f,t) \in I^E$ and equal to zero otherwise,

$Y_{f_1,f_2} \geqslant 0 \quad \forall (f_1,f_2) \in Q$ and equal to zero otherwise.

$H_{c,f} \in [0,1] \quad \forall c \quad$ and $\quad f \in L_{34} \cap A_c$,

$Z_{c,t} \geqslant 0 \quad \forall c, t,$

$P_{t,j} \geqslant 0 \quad \forall t, j = 1, \ldots, n,$

$W_{c,t} \geqslant 0 \quad \forall c, t.$

Remark 3. (A) Given binary values for the X-variables, integrality of the Y, Z, P, and W-variables is automatically implied at (alternative optimal) extreme points of the resultant feasible region.

(B) Constraints (EC_2)–($\mathrm{EC}_{9.2}$) of Model ECFAM are, respectively, similar to constraints (C_2)–($\mathrm{C}_{9.2}$) of Model CFAM, except that the restriction $c \in C_t$ is now discarded in all the constraints of Model ECFAM.

6. Gender policy modeling considerations

Model ECFAM described above does not explicitly identify male, female, or joint sections of offered classes. Section 6.1 accordingly investigates restrictions related to teaching male, female, and joint class sections and Section 6.2 examines the issue of offering patterns related to multiple-section classes.

6.1. Teaching restrictions

Based on the model ECFAM, a female or male faculty member can teach a male, female, or joint section. This model can be modified to assign, to the extent possible, male sections to male faculty members, and female sections to female faculty members, while joint sections may be taught by either female or male faculty members. Currently, the number of female students is considerably higher than the number of male students, and consequently, the number of female sections of a given class is proportionately greater than the number of male sections of that class.

Assume that all faculty members can teach female sections. Some female faculty members strongly prefer teaching female sections. The number of such faculty members is relatively very small as compared with the overall number of faculty members. This requirement can be thus guaranteed by the following constraint, where $MJ \subseteq L$ is composed of all faculty members who are willing to teach male and joint sections.

$$\sum_{f \in C^f \cap MJ} X_{c,f,t} \geqslant M_{c,t} + J_{c,t}.$$

6.2. Offering patterns

Recall that D_c is the targeted maximum number of sections that may be offered during any time-slot t for a multiple-section class c. For better offering patterns, we consider the following two cases.

6.3. Classes having two sections

Currently, such classes are offered as joint sections. Hence, if c is a two-section class, then it is better to offer the two sections of c during different time-slots, and thus a suitable value for D_c is 1. On the other hand, such sections may be offered as (a) male and female sections, (b) male and joint sections, or (c), female and joint sections. For Case (a), a suitable value for D_c is 2 while for (b) and (c), a suitable value for D_c is 1.

6.4. Classes having more than two sections

As mentioned earlier, there are more female students than male students, where $F_c \geqslant M_c$ for almost all c. Therefore, it is reasonable to allow, to the extent possible, two sections of a given class to be offered during any time-slot only if one section is allocated for female students while the other section is allocated for male students or is a joint section. Accordingly, if c is such a class, then a suitable value for D_c is 2. Note that this choice of D_c may also lead to solutions in which two female sections or two joint sections are offered during the same time-slot; nonetheless, this is still acceptable since in most cases, students have sufficient time-slot alternatives for such classes.

Remark 4. If it is necessary to attain some desirable restrictive offering pattern for c, we can use $D_{c,t}$ instead of D_c, where $D_{c,t}$ denotes the targeted maximum number of sections of c that may be offered during a given time-slot t.

7. Solution algorithms and computational results

In this section, we present solution algorithms, computational results, and sensitivity analyses for the model ECFAM based on actual class and faculty data for six academic terms. Section 7.1 presents two heu-

ristics to solve Model ECFAM. A what-if analysis based on comparisons of schedules generated via the proposed approach with the manual approach is presented in Section 7.1. Note that the mathematics classes offered by the Department of Mathematics and Computer Science can be assigned to faculty members separately from the computer science classes offered by the same department. This is so because in most cases, a faculty member in the mathematics group teaches only mathematics classes and a faculty member in the computer science group teaches only computer science classes. The number of mathematics classes offered during a term is significantly greater than the number of computer science classes, and therefore, we focus on the mathematics classes because the corresponding ECFAM model is more challenging to solve.

We present results related to six academic terms for the mathematics classes at the Department of Mathematics and Computer Science at Kuwait University, for which F99-0, F00-1, and F01-2, respectively denote the fall terms for the academic years 1999–2000, 2000–2001, and 2001–2002, and S01-2, S02-3, and S03-4, respectively denote the spring terms for the academic years 2001–2002, 2002–2003, and 2003–2004. Detailed statistics related to these instances (which involve up to 30 different classes, 94 class sections, and 37 faculty members) can be gleaned at www.al-yakoob.com.

Next, we define a glossary of notation that will be used in this section.

- $\bar{P}$: linear relaxation of any problem P.
- $v(P)$: optimal objective function value of any problem P.
- $v_{\text{ub}}(P)$: best solution (upper bound) found for problem P.
- $v_{\text{lb}}(P)$: best lower bound found for problem P.
- $\alpha = \sum_{(c,f,t)\in I^E} DC(c,f,t)X_{c,f,t}$.
- $\beta = \mu_1 \sum_{(f_1,f_2)\in Q} Y_{f_1,f_2}$.
- α^m: the summation of the individual dissatisfaction levels for all faculty members when the class schedule is generated manually.
- β^m: the summation of all the absolute differences between the dissatisfaction levels of faculty members having the same load, when the class schedule is generated manually.
- $\mu^m(l)$: the mean for the individual dissatisfaction levels obtained via the manual approach for faculty members having class load $l \in \{1,2,3\}$.
- $\mu^{\text{SFVH}}(l)$: the mean for the individual dissatisfaction levels obtained via Procedure SFVH for faculty members having class load $l \in \{1,2,3\}$.
- $\sigma^m(l)$: the standard deviation for the individual dissatisfaction levels obtained via the manual approach for faculty members having class load $l \in \{1,2,3\}$.
- $\sigma^{\text{SFVH}}(l)$: the standard deviation for the individual dissatisfaction levels obtained via Procedure SFVH for faculty members having class load $l \in \{1,2,3\}$.
- opt-gap: optimality gap.
- RT: run time in CPU seconds on a Pentium IV computer having 256 MB of RAM and using CPLEX-7.5, where all procedures have been coded in C.
- $|S|$: Number of elements in any set S.

7.1. LP-based heuristic procedures for solving Model ECFAM

Recall that Model ECFAM is a modified version of CFAM that attempts to enhance generated faculty schedules, however, with a limit on the number of allowable changes as discussed in Section 5. Accordingly Model ECFAM involves a considerably greater number of binary variables than Model CFAM, and hence, the former is likely to be harder to solve. Thus, the emphasis of the remainder of this section will be on devising effective solution algorithms for Model ECFAM. This is further justified by noting that setting TAC = 0 in ECFAM, reduces this model to an instance of Model CFAM.

Initially, we attempted to solve Model ECFAM directly via CPLEX based on the six test problems. For all test problems, we ran CPLEX until we encountered out-of-memory difficulties. The best performance was for Test Problem S03-04, and for that too, the solution obtained was only within 50% of optimality (i.e., the optimality gap obtained was about 0.5). The time needed to obtain this solution was slightly more than two days. For all the other test problems, we encountered out-of-memory difficulties before reaching any solution that was within 50% of optimality. Consequently, we first attempted to design a two-phase LP-based heuristic in which $\overline{\text{ECFAM}}$ is solved, and its solution is used to determine the number of sections of each class to be offered during any given time-slot. Modifying the values of $O_{c,t}$ accordingly, and setting $\text{TAC} = 0$, we then solve Model ECFAM, essentially as an instance of CFAM.

The best performance using this heuristic was again obtained for Test Problem S03–04, which yielded a solution within 68% of optimality (i.e., the optimality gap was about 0.32). The time needed to obtain this solution was still more than two days. Again, for all the other test problems, we encountered out-of-memory difficulties before reaching meaningful solutions. Hence, we designed the following alternative heuristic method that we propose for implementation.

The out-of-memory difficulties associated with solving Model ECFAM primarily stems from the large number of binary variables. This motivates the development of a heuristic procedure that generates a solution in an iterative fashion, in which each iteration involves a modified version of Model ECFAM where the integrality of only a subset of the integer variables is enforced. This heuristic procedure is described next.

In Model ECFAM, let the vector X be partitioned as $X = (X_1, X_2, \ldots, X_C)$, where for $c \in \{1, \ldots, C\}$, X_c denotes the vector of X-variables associated with class c. Partition the set $\{1, \ldots, C\}$ into $\omega \geqslant 1$ partitions, denoted by $\Phi_1, \ldots, \Phi_\omega$. Hence, $\bigcup_{j=1}^{\omega} \Phi_j = \{1, \ldots, C\}$ and $\Phi_j \cap \Phi_k = \emptyset$, $\forall j, k \in \{1, \ldots, \omega\}$, and $j \neq k$. Furthermore, we will let $\overline{\text{ECFAM}}$ (1, opt-gap^1) denote the relaxation of Model ECFAM for which integrality is enforced only on the X-variables that correspond to $c \in \Phi_1$ and an optimal solution is required to be found within a tolerance "opt-gap^1" of optimality. For $r = 2, \ldots, \omega$, we define $\overline{\text{ECFAM}}$ (r, opt-gap^r) to be the linear relaxation of model ECFAM having the following characteristics:

(a) The values of all the X-variables that correspond to $\bar{c} \in \bigcup_{j=1}^{r-1} \Phi_j$ are fixed as obtained from $\overline{\text{ECFAM}}$ (j, opt-gap^j), for $j = 1, \ldots, r-1$.
(b) The integrality on the X-variables that correspond to $c \in \Phi_r$ is enforced.
(c) An optimal solution is required to be found within a tolerance "opt-gap^r" of optimality.

The sequential fixing of variables heuristic SFVH then proceeds as follows.

Initialization. Let ω be some integer number less than or equal C. Let $r = 1$ and let opt-gap^1 be some selected optimality gap criterion. Solve Model $\overline{\text{ECFAM}}$ (1, opt-gap^1).

Main Step: If $r = \omega$; then terminate the algorithm; the proposed solution is that obtained from solving Model $\overline{\text{ECFAM}}$(ω, opt-gap$^\omega$). Otherwise, increment r by one and solve the model $\overline{\text{ECFAM}}$(r, opt-gap^r). Repeat the Main Step.

A partitioning scheme for $X = (X_1, X_2, \ldots, X_C)$

The actual value of ω depends on the problem data, the required overall optimality of solution, and the maximum time-frame during which a solution should be obtained. In general, smaller values of ω should produce better quality solutions. Note that based on our extensive computational testing and analysis, we observed that in some cases enforcing the integrality of one extra variable changes the solvability of the problem. Also, for the six test problems, judicial values of ω were found to be 3 or 4, when the maximum run time for any iteration was set to 1 hour. Furthermore, it seems reasonable to set Φ_1 to be the set of one- and two-section classes to the extent possible. This stems from the fact that class conflicts mostly involve

only one- or two-section classes, in which case, it is better to determine the time-slots for such classes first, and then specify the time-slots for the multiple section classes.

7.2. Lower and upper bounds

Recall that $\overline{\text{ECFAM}}(\omega, \text{opt-gap}^{\omega})$ yields an overall feasible solution whenever it produces an optimum, and hence, the objective function value of this problem constitutes the best available upper bound. Moreover, the objective function value of $\overline{\text{ECFAM}}(1, \text{opt-gap}^{1})$ gives a (best available) lower bound on the overall problem. Hence, $v_{\text{ub}}(\text{ECFAM}) = v(\overline{\text{ECFAM}}(\omega, \text{opt-gap}^{\omega}))$ and $v_{\text{lb}}(\text{ECFAM}) = v(\overline{\text{ECFAM}}(1, \text{opt-gap}^{1}))$.

Let $v(\text{ECFAM}_r) = \overline{\text{ECFAM}}(r, \text{opt-gap}^{r})$, for $r = 1, \ldots, \omega$, and let perct_opt $(v_{\text{ub}}, v_{\text{lb}}) = 100\left(1 - \frac{v_{\text{ub}}(\text{ECFAM}) - v_{\text{lb}}(\text{ECFAM})}{v_{\text{ub}}(\text{ECFAM})}\right)$. Tables 1–3 present statistics related to solving Model ECFAM for the six test problems using Procedure SFVH, where $\text{opt-gap}^{r} = 0.10$ for $r = 1, \ldots, \omega$.

Note that the linear relaxation of Model ECFAM was readily solved for all test problems as illustrated in Table 1. Also, Table 3 indicates that Procedure SFVH produced solutions having optimality percentages ranging from 55% to 73%. For terms F99-0 and F03-4, solutions were obtained using three iterations and for the other terms, four iterations were needed to reach the stated solutions.

Table 1
Statistics related to ECFAM

Term	Rows	Columns	Nonzero entries	v ($\overline{\text{ECFAM}}$)	RT
F99-0	2273	9622	435,444	630.51	1.77
F00-1	1843	7623	280,886	662.76	2.14
F01-2	1737	8323	264,576	545.21	2.37
S01-2	2003	9626	342,001	619.09	2.65
S02-3	2040	9033	398,053	614.37	2.56
F03-4	1727	6420	23,551	523.31	2.07

Table 2
Statistics related to solving ECFAM using procedure SFVH

Term	v (ECFAM_1)	RT	v (ECFAM_2)	RT	v (ECFAM_3)	RT
F99-0	689.04	185.00	789.71	2.00	1080.00	345.00
F00-1	703.00	1577.00	819.83	2.00	975.01	2.00
F01-2	615.69	980.94	67 8.69	2.00	733.60	1.00
S01-2	682.00	47,060.70	881.00	2230.00	898.00	1.00
S02-3	677.66	66.00	753.35	33.00	971.68	1.00
F03-4	709.00	133.00	794.40	53.00	983.00	16.00

Table 3
Statistics related to solving ECFAM using procedure SFVH (continued)

Term	v (ECFAM_4)	RT	v_{ub} (CFAM)	Total RT	perct_opt (v_{ub}, v_{lb})
F99-0	N/A	N/A	1080.00	532.00	64
F00-1	1066.00	2.00	1066.00	1583.50	66
F01-2	914.00	1.30	914.00	985.24	68
S01-2	1070.00	31.08	1070.00	49,323.58	64
S02-3	1247.00	2.50	1247.00	102.50	55
F03-4	N/A	N/A	983.00	150.20	73

Table 4
Improvement of proposed modeling approach over the manual approach

Term	$100\left(\frac{\alpha^m-\alpha}{\alpha^m}\right)$	$100\left(\frac{\beta^m-\beta}{\beta^m}\right)$	$100\left(\frac{(\alpha^m+\beta^m)-(\alpha+\beta)}{\alpha^m+\beta^m}\right)$
F99-0	57.20	92.20	85.0
F00-1	57.60	94.70	86.5
F01-2	59.60	94.3	87.6
S01-2	59.50	95.70	89.8
S02-3	57.00	92.20	85.0
F03-4	45.11	83.18	72.75

Table 5
Means and standard deviations of the dissatisfaction levels obtained manually and via SFVH

Term	$\mu^m(l)$			$\mu^{\mathrm{SFVH}}(l)$			$\sigma^m(l)$			$\sigma^{\mathrm{SFVH}}(l)$		
	l			l			l			l		
	1	2	3	1	2	3	1	2	3	1	2	3
F99-0	22.50	37.61	50.00	13.50	9.14	19.92	0.70	16.28	12.06	2.12	2.19	4.01
F00-1	20.66	41.00	54.63	7.66	11.00	20.20	7.57	15.59	15.79	5.03	3.46	3.70
F01-2	18.00	38.66	55.70	4.50	13.75	20.00	8.48	15.81	20.56	2.12	1.28	0.86
S01-2	17.33	42.88	56.04	3.00	13.14	23.26	11.01	14.65	24.41	1.73	3.07	2.95
S02-3	15.00	45.70	56.11	5.00	11.57	18.60	16.47	18.83	19.21	1	3.15	5.59
F03-4	16.16	39.16	55.53	4.50	17.72	23.94	9.84	18.70	16.11	8.09	0.90	2.73

7.3. Comparison of dissatisfaction levels with the current and proposed assignment approaches

This section presents a comparison of the dissatisfaction levels with respect to the schedules generated via the current manual and the proposed modeling approach. The results in Tables 4 and 5 reveal that the modeling approach by far outperforms the current manual approach from both an individual and fairness (or equity) points of view. Furthermore, not only are the mean dissatisfaction levels higher using the manual approach, but the manual approach also produces a greater disparity among faculty members, as evidenced by the values of $100\left(\frac{\beta^m-\beta}{\beta^m}\right)\%$ in Table 4, and the use $\sigma^m(l)$ and $\sigma^{\mathrm{SFVH}}(l)$ in Table 5.

8. Summary, conclusions, and future research

This paper addresses a faculty-class assignment problem, which represents Stage II of a two-stage approach that the authors are adopting to tackle the class-scheduling problem at Kuwait University. Two mixed-integer programs, denoted by CFAM and ECFAM, are developed in this paper to assign faculty members to specific classes and time-slots. The model CFAM generates class schedules that assign faculty members to female, male, and joint sections of offered classes, taking into consideration faculty members' preferences for specific classes and time-slots. The enhanced model ECFAM further attempts to improve the overall satisfaction of faculty members by allowing some changes in time-slots for course offerings as permitted by the Office of Registration while maintaining certain essential features related to class conflicts and efficient facility utilization. Moreover, modeling strategies are discussed in order to maintain desirable class offering patterns under imposed gender-based policies. Solving Model ECFAM via a direct application of the commercial package CPLEX-MIP (version 7.5) was out-of-reach for the six test problems based

on a case study related to the Department of Mathematics and Computer Science at Kuwait University. This motivated the development of an iterative heuristic procedure in which the integrality of only a subset of the integer variables is enforced at any given iteration. The proposed heuristic generated good quality solutions in reasonable times for all the test problems. Manually generated faculty-class schedules are often biased and require extensive effort. However, the schedules generated via the proposed approach can be obtained in a timely fashion, and the results indicate that the overall satisfaction levels of faculty members can be noticeably enhanced and made significantly more equitable by using this method, as compared with those obtained using the existing manual schedule generation procedure. Alternative mathematical formulations for the class-faculty assignment problem addressed in this paper, which are amenable to be solved using column generation approaches, are developed and tested in a companion paper [1]. Finally, the models presented in this paper and in [1] for the Stage II problem along with those developed in [2] for the Stage I problem will be incorporated, within a decision support system (DDS) to replace the existing manual approach at the Department of Mathematics and Computer Science at Kuwait University. This DDS will then be expanded to incorporate other colleges in the university.

Acknowledgements

This research is supported by *Kuwait Foundation for the Advancement of Sciences* under Grant No [2003-15-01] and the *National Science Foundation* under Grant No. [DMI-0094462]. Special thanks to Mrs. Basha'er Al-Qazwini for her contributions to the computational implementation of the algorithms.

Appendix A. Summary of notation

The following is a summary of notation presented here for the sake of ease in reference.

- $t = 1, \ldots, A$: indices for the SMW time-slots.
- $t = A + 1, \ldots, A + B$: indices for the ST time-slots.
- $T_{\mathrm{SMW}} = \{1, \ldots, A\}$.
- $T_{\mathrm{ST}} = \{A + 1, \ldots, A + B\}$.
- $T = \{1, \ldots, A + B\}$.
- $c = 1, \ldots, C$: different classes that are offered during the specified term.
- $F_{c,t}$, $M_{c,t}$, and $J_{c,t}$: respectively, denote the number of female, male, and joint sections of class c that are offered during time-slot t.
- F_c, M_c, and J_c: respectively denote the numbers of female, male, and joint sections of class c, where $F_c = \sum_t F_{c,t}, M_c = \sum_t M_{c,t}$, and $J_c = \sum_t J_{c,t}$.
- $E_c = (F_c, M_c, J_c)$.
- $O_{c,t} = F_{c,t} + M_{c,t} + J_{c,t}$: the total number of sections of class c that are offered during time-slot t, regardless of gender.
- $O_c = \sum_t O_{c,t}$: the total number of sections of class c that are offered during the different time-slots.
- $O_t = \sum_c O_{c,t}$: the total number of sections of the different classes that are offered during time-slot t.
- $s_1, s_2, \ldots, s_m$: all possible designated class sizes.
- $S_1, S_2, \ldots, S_m$: for $j = 1, \ldots, m$, S_j is composed of all classes having class size s_j.
- R_{t,s_j}: the number of classes of size s_j that are held during time-slot t.
- $f = 1, \ldots, F$: index the faculty members that are available for the term.
- L^f: class-load for faculty member f.
- L_1, L_2 and L_3: the sets of faculty members teaching one, two, and three classes, respectively.

- L_{21}: the set of faculty members in L_2 who are willing to teach two classes consecutively.
- L_{22}: the complement of L_{21} in L_2.
- L_{31}: the set of faculty members in L_3 who are willing to teach two classes consecutively.
- L_{32}: the complement of L_{31} in L_3.
- L_{33}: the set of faculty members in L_3 who are willing to teach three different classes.
- L_{34}: the complement of L_{33} in L_3.
- $L_3 = L_{31} \cup L_{32} = L_{33} \cup L_{34}$. $L = L_1 \cup L_2 \cup L_3$.
- $Q_i = \{(f_1, f_2)$: where both f_1 and f_2 belong to L_i with $f_1 < f_2\}$, $i = 1, 2, 3$.
- $Q = Q_1 \cup Q_2 \cup Q_3$.
- $MJ \subseteq L$: the set of faculty members who are willing to teach male and joint sections of offered classes.
- T_c: the set of time-slots during which a section of class c is offered.
- A_c: the set of faculty members that can teach class c.
- C^f: the set of classes that can be taught by faculty member f.
- C_t: the set of classes that are offered during time-slot t.
- $I = \{(c,f,t) : c \in \{1, \ldots, C\}, f \in A_c$, and $t \in T_c\}$.
- $I^E = \{(c,f,t) : c \in \{1, \ldots, C\}, f \in A_c$, and $t \in T\}$.
- $\kappa(c,f)$: the priority associated with class c as indicated by faculty member f.
- $\lambda(f,t)$: the priority associated with time-slot t as indicated by faculty member f.
- $\mathrm{DC}(c,f,t) = \kappa(c,f) + \lambda(f,t)$.
- TAC: the total number of allowable changes in a given term.
- D_c: the maximum number of sections that may be offered during any time-slot t for a multiple-section class c.
- $D_{c,t}$: the maximum number of sections of c that may be offered during a given time-slot t.

References

[1] S.M. Al-Yakoob, H.D. Sherali, A column generation approach for a class-faculty assignment problem. Working paper, Department of Mathematics and Computer Science, College of Science, Kuwait University, Kuwait, 2004.

[2] S.M. Al-Yakoob, H.D. Sherali, A class scheduling and timetabling problem with gender policies and traffic considerations. Working paper, Department of Mathematics and Computer Science, College of Science, Kuwait University, Kuwait, 2004.

[3] S.M. Al-Yakoob, H.D. Sherali, Mixed-integer programming models for an employee scheduling problem with multiple shifts and work locations. Working paper, Department of Mathematics and Computer Science, College of Science, Kuwait University, Kuwait, 2004.

[4] A.S. Asratian, D. deWerra, A generalized class–teacher model for some timetabling problems, European Journal of Operational Research 143 (2002) 531–542.

[5] A.S. Asratian, N.N. Mirumian, Transformations of school timetables, Matematicheskie Voprosy Kibernetiki 4 (1992) 93–110 (in Russian).

[6] J. Aubin, J. Ferland, A large scale timetabling problem, Computers and Operations Research 16 (1) (1989) 67–77.

[7] D. Costa, A Tabu search algorithm for computing an operational timetable, European Journal of Operational Research 76 (1994) 98–110.

[8] D. de Werra, An introduction to timetabling, European Journal of Operational Research 19 (1985) 151–162.

[9] D. de Werra, Extensions of coloring models for scheduling purposes, European Journal of Operational Research 92 (1996) 474–492.

[10] K. Dowsland, A timetabling problem in which clashes are inevitable, Journal of the Operational Research Society 41 (10) (1990) 907–918.

[11] R. Fahrion, G. Dollansky, Construction of university timetables using logic programming techniques, Discrete Applied Mathematics 35 (1992) 221–236.

[12] J. Ferland, C. Fleurent, A decision support system for course scheduling, Interfaces 24 (1994) 105–115.

[13] I. Giannikos, El-Darazi, P. Lees, An integer goal programming model to allocate offices to staff in an academic institution, Journal of the Operational Research Society 46 (6) (1995) 713–720.

[14] K.C. Gilbert, R.B. Hofstra, An algorithm for a class of three-dimensional assignment problems arising in scheduling applications, IIE Transactions (March) (1987) 29–33.
[15] C.C. Gotlieb, The construction of class–teacher timetables, in: Processing IFIP Congress, Amsterdam, The Netherlands, 1963, pp. 73–74.
[16] N.L. Lawrie, An integer programming model of a school timetabling problem, The Computer Journal 12 (1969) 307–316.
[17] R. Nada, J. Browne, Introduction to Employee Scheduling, Van Nostrand Reinhold, New York, NY, 1992.
[18] I. Ozkarahan, J.E. Bailey, Goal programming model subsystem of a flexible nurse scheduling support system, IIE Transactions 20 (3) (1988) 306–316.
[19] I. Ozkarahan, Flexible hospital resource scheduling model, In R. J. Ebert, L. Franz (Eds.) Proceedings of the 27th Annual Meeting of the Decision Sciences Institute, vol. 2(2) Orlando, FL, November 24–26, 1996, pp. 1030–1032.
[20] A. Tripathy, School timetabling—a case in large binary integer linear programming, Management Science 30 (12) (1984) 13–19.
[21] K. Yura, Production scheduling to satisfy worker's preferences for days off and overtime under due-date constraints, International Journal of Production Economics 33 (1–2) (1994) 265–270.